

WAN Architectures

- Topics covered:
 - Intro To WANs
 - Leased Lines
 - MPLS VPNs
 - Internet Connectivity
 - Internet VPNs

Intro To WANs

WAN Architecture?

WAN stands for **Wide Area Network**.

A WAN is a network that extends over a large geographic area, such as Between cities, Between countries.

WANs are used to connect geographically separate LANs (Local Area Networks).

For example, if a company has offices in:

- New York
- Toronto
- London

Each office is its own LAN, and the connections between them form a WAN.

Although the Internet itself can be considered a WAN, the term **WAN** is usually used to describe an enterprise's private connections that connect:

1. Offices
2. Data centers
3. Other sites

together.

VPNs Over Public Networks

Public networks such as the Internet are shared by many users.

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To create private connections over public networks, companies use VPNs (Virtual Private Networks).

VPNs allow private communication over shared infrastructure.

Different VPN technologies can be used to create secure private connections between sites.

WAN Technologies

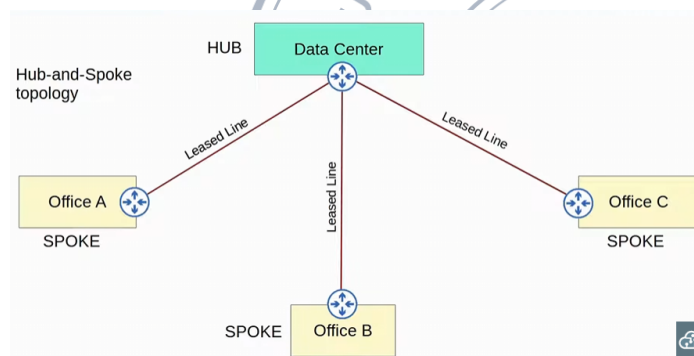
Many WAN technologies have existed over the years.

1. Some technologies are available in certain locations and unavailable in others.
2. Technologies considered legacy in one country may still be used elsewhere.
3. Legacy means a technology that is old and no longer commonly used.

Example WAN Using Leased Lines

Each office and the data center are separate LANs.

The offices connect to the data center through leased lines.



Leased lines are:

- Dedicated physical connections
- Used between sites
- Private connections
- Not shared with the Internet

Hub-and-Spoke Topology

This WAN design is called a Hub-and-Spoke Topology.

Components

- Hub: Central site (Data Center)
- Spokes: Remote offices connected to the hub

Advantage

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Compared to a full-mesh topology, hub-and-spoke makes it easier to centrally control traffic.

For example:

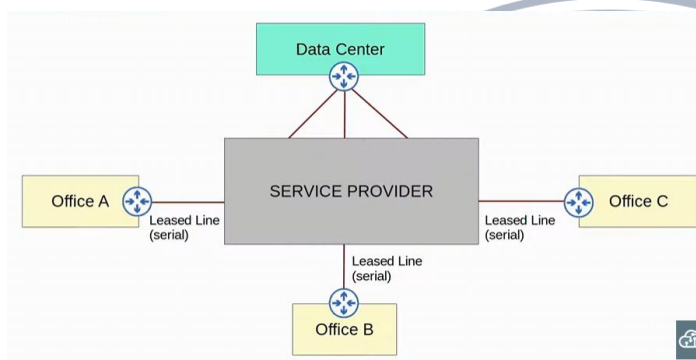
- Traffic between offices can pass through a firewall at the data center.
- The firewall controls which traffic is permitted and which is not.

Remember the term: **Hub-and-Spoke Topology**

More Accurate Representation of Leased Lines

The previous diagram is simplified.

In reality, sites connect through a service provider.



Instead of one cable directly connecting each site, every site connects to the service provider, which connects the sites together.

Leased Line Characteristics

Leased lines use:

- Serial connections
- PPP encapsulation
- HDLC encapsulation

These are not Ethernet links.

Layer 2 encapsulation is: PPP, HDLC rather than Ethernet.

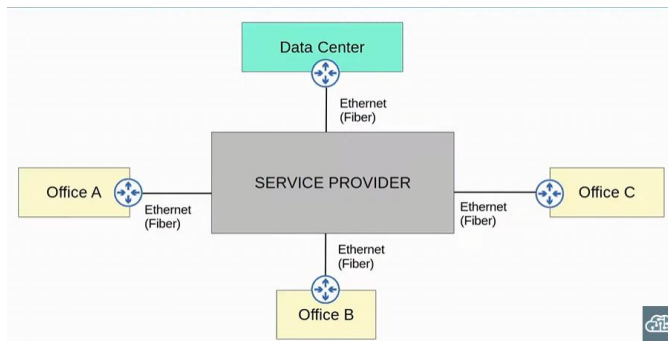
Ethernet WAN Connections

Modern WANs increasingly use Ethernet.

Optical fiber allows much longer distances than traditional copper UTP Ethernet cables.

Because of this, Ethernet WANs using fiber optic connections have become common.

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Enterprise Perspective

The CCNA focuses on WANs from the enterprise perspective.

The internal operation of the service provider network is not a focus.

The service provider network is represented as the gray box in the diagrams.

Internet-Based WAN Connections

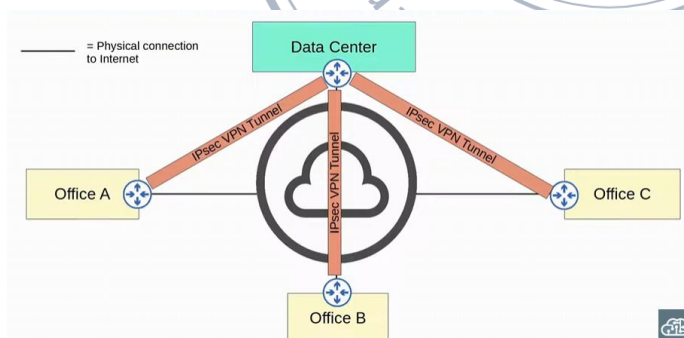
The Internet can also be used to connect enterprise sites.

However:

- The Internet is public.
- The Internet is shared.
- Sending important data without protection is not desirable.

In this design, each site has a physical Internet connection.

VPNs are used to securely transfer data between sites.



Physical Internet connections exist at each site.

VPN Operation

1. Packets are encrypted.
2. Only intended recipients can read the contents.
3. The encrypted packet is encapsulated inside a new packet.
4. The packet is then sent across the Internet.
5. The original packet remains protected while traveling across the public network.

Leased Lines

What is Leased Line?

A leased line is:

- A dedicated physical link
- Typically connecting two sites

Characteristics:

- Uses serial connections
- Uses PPP or HDLC encapsulation
- Does not use Ethernet encapsulation

Leased Line Standards

Different countries use different leased line standards and speeds.

The chart shown includes several standards and data rates.

System	North American	Japanese	European (CEPT)
Level zero (channel data rate)	64 kbit/s (DS0)	64 kbit/s	64 kbit/s
First level	1.544 Mbit/s (DS1) (24 user channels) (T1)	1.544 Mbit/s (24 user channels)	2.048 Mbit/s (32 user channels) (E1)
(Intermediate level, T-carrier hierarchy only)	3.152 Mbit/s (DS1C) (48 Ch.)	–	–
Second level	6.312 Mbit/s (DS2) (96 Ch.) (T2)	6.312 Mbit/s (96 Ch.), or 7.786 Mbit/s (120 Ch.)	8.448 Mbit/s (128 Ch.) (E2)
Third level	44.736 Mbit/s (DS3) (672 Ch.) (T3)	32.064 Mbit/s (480 Ch.)	34.368 Mbit/s (512 Ch.) (E3)
Fourth level	274.176 Mbit/s (DS4) (4032 Ch.)	97.728 Mbit/s (1440 Ch.)	139.264 Mbit/s (2048 Ch.) (E4)
Fifth level	400.352 Mbit/s (DS5) (5760 Ch.)	565.148 Mbit/s (8192 Ch.)	565.148 Mbit/s (8192 Ch.) (E5)

Wikipedia: 'Comparison of T-carrier and E-carrier systems'

North American Standards: Naming begins with T1, T2, T3

These correspond to different transmission speeds.

European Standards: Naming begins with E1, E2, E3

These also correspond to different transmission speeds.

Why Ethernet WANs Are Becoming More Popular

Compared to Ethernet WAN technologies, leased lines generally have:

1. Higher cost
2. Longer installation lead time
3. Slower speeds

Because of these factors, Ethernet WAN technologies are becoming increasingly popular.

MPLS VPNs

MPLS (Multiprotocol Label Switching)

MPLS stands for **Multiprotocol Label Switching**.

Like the Internet, an MPLS network is shared infrastructure because multiple customer enterprises connect to and use the same service provider network.

However, MPLS allows VPNs (Virtual Private Networks) to be created over this shared infrastructure through the use of **labels**.

These labels:

- Separate different customers' traffic.
- Prevent traffic from mixing with traffic belonging to other customers.
- Allow private communication over shared infrastructure.

MPLS Terminology

There are three important router types:

CE Router (Customer Edge)

- Customer-owned router.
- Located at the edge of the customer network.

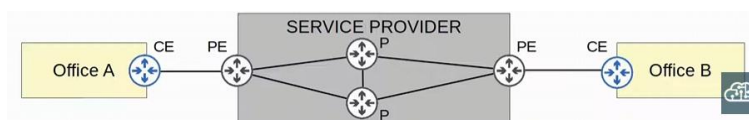
PE Router (Provider Edge)

- Service provider router.
- Connects directly to customer CE routers.

P Router (Provider Router)

- Core router within the service provider network.
- Does not connect directly to customer routers.

MPLS Network Structure



1. CE routers sit at the edge of customer networks.
2. CE routers connect to PE routers.
3. P routers form the provider's internal network infrastructure.
4. P routers do not directly connect to customer routers.

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How MPLS Forwards Traffic

When a PE router receives a frame from a CE router:

1. It adds an MPLS label.
2. The label is inserted between:
 - Layer 2 Ethernet Header
 - Layer 3 IP Header

Because the label exists between Layer 2 and Layer 3, MPLS is sometimes called a:

Layer 2.5 Protocol

MPLS Forwarding Decisions

In traditional IP routing:

- Routers examine the destination IP address.
- The routing table is consulted.
- A forwarding decision is made.

In MPLS:

- Routers do not use the destination IP for forwarding decisions.
- MPLS labels are used instead.

The MPLS label determines where traffic is forwarded inside the provider network.

CE Routers and MPLS

CE routers:

1. Do not use MPLS.
2. Do not need to run MPLS.
3. Do not even need MPLS capability.

Only:

- PE routers
- P routers

use MPLS.

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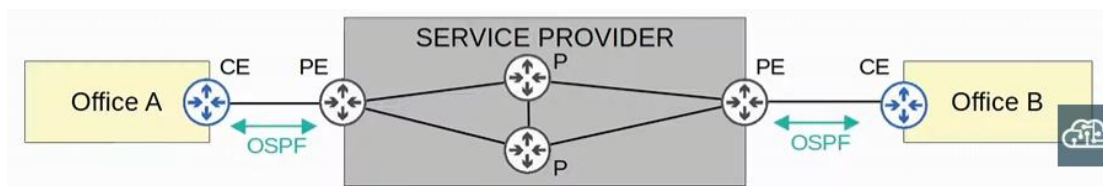
Layer 3 MPLS VPN

One type of MPLS VPN is a Layer 3 MPLS VPN.

In this design:

- CE routers and PE routers form routing protocol peerings.
- OSPF is used as an example.
- Other routing protocols can also be used.
- Static routes can also be configured with PE routers as the next hop.

Layer 3 MPLS VPN Structure



Operation

1. Office A's CE peers with a PE router.
2. Office B's CE peers with another PE router.
3. Routing information is exchanged through those peerings.
4. Office A learns Office B routes.
5. Office B learns Office A routes.

In a Layer 3 MPLS VPN:

- CE routers peer with PE routers.
- Dynamic routing protocols or static routes are used.

Layer 2 MPLS VPN

A Layer 2 MPLS VPN works differently.

In this design:

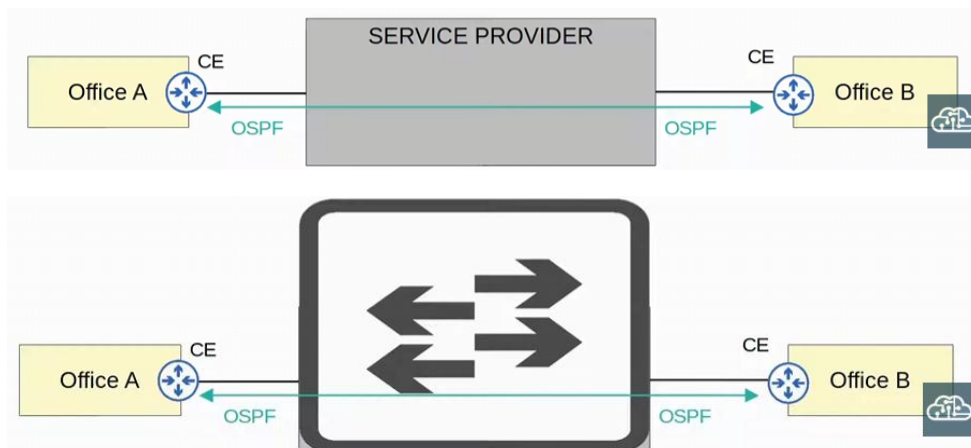
- CE routers do not form peerings with PE routers.
- The service provider network becomes transparent to CE routers.

Although CE routers physically connect to PE routers, it appears as though the CE routers are directly connected to each other.

Their WAN interfaces are in the same subnet.

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Layer 2 MPLS VPN Structure



Operation

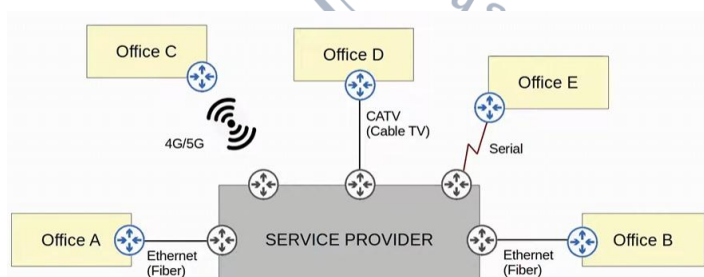
1. Routing protocols run directly between CE routers.
2. The provider network behaves like one large switch.
3. CE routers communicate as if directly connected.

Key Point

The provider network still runs MPLS internally, but from the customer perspective it behaves like a large Layer 2 switch.

MPLS Access Technologies

MPLS operates within the service provider network, but different connection types can be used to access that MPLS infrastructure.



Connection Types Shown

1. Office A → Ethernet (Fiber)
2. Office B → Ethernet (Fiber)
3. Office C → 4G/5G Wireless
4. Office D → CATV (Cable TV)
5. Office E → Serial leased line

All sites can communicate through the service provider's MPLS infrastructure.

Internet Connectivity

Internet Connectivity in Detail

There are many ways for an enterprise to connect to the Internet.

Examples include:

1. Leased lines
2. MPLS VPNs
3. CATV
4. DSL
5. Fiber optic Ethernet

Although leased lines and MPLS VPNs are private technologies, they can be used to access the public Internet.

Fiber optic Ethernet continues to become more popular because it provides high speeds over long distances.

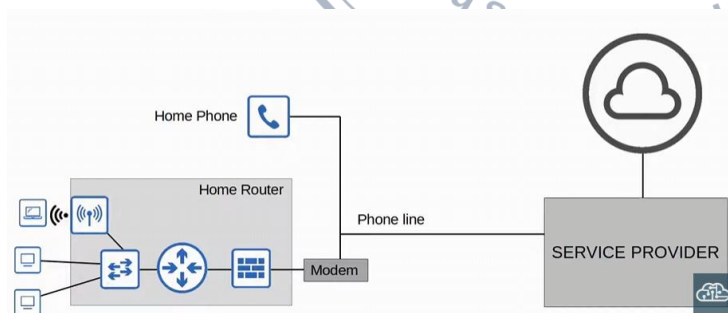
DSL (Digital Subscriber Line)

DSL stands for Digital Subscriber Line.

DSL provides Internet access using existing telephone lines.

This is convenient because the phone line is already installed in most homes.

DSL Structure



DSL Modem

A modem is required.

Modem stands for: **Modulator-Demodulator**

Purpose: Converts data into a format suitable for transmission over phone lines.

The modem may be: A separate device and Built into the home router

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Cable Internet

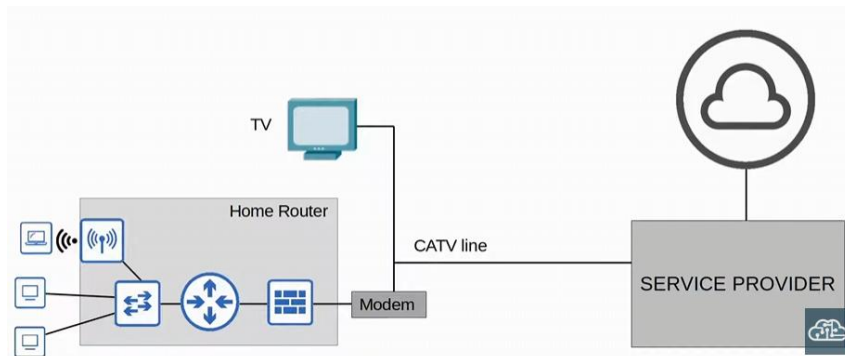
Cable Internet is similar to DSL.

Instead of phone lines, it uses:

CATV (Cable Television) lines

The same cable lines used for television service provide Internet access.

Cable Internet Structure



Cable Modem

A cable modem:

- Converts data into a format suitable for transmission over CATV cables.
- Can be a separate device.
- Can be integrated into the home router.

Internet Connection Redundancy

For enterprises, Internet access is often essential.

Redundant Internet connections improve availability.

There are four important terms.

1. Single-Homed

One connection to one ISP.

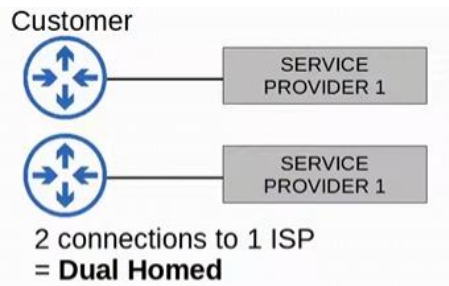
Customer ----- ISP 1

- Standard home Internet model.
- No redundancy.

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2. Dual-Homed

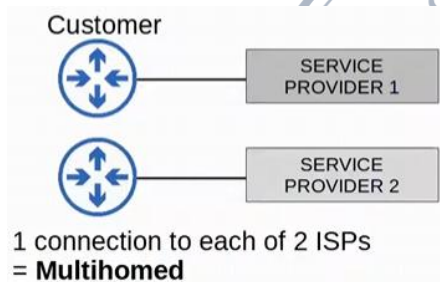
Two connections to one ISP.



- Provides some redundancy.
- Still relies on a single ISP.

3. Multihomed

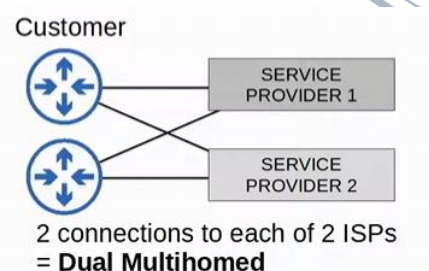
One connection to each of two ISPs.



- Better redundancy.
- Internet access remains available if one ISP has a problem.

4. Dual Multihomed

Two connections to each of two ISPs.



- Provides the highest redundancy.
- May not always be necessary or cost-effective.

Internet VPNs

Introduction To Internet VPNs

Private WAN technologies such as: Leased Lines, MPLS provide security because customer traffic is separated either by:

- Dedicated physical connections (Leased Lines)
- MPLS labels/tags (MPLS)

However, when using the Internet as a WAN, there is no built-in security by default.

To provide secure communication over a shared network such as the Internet, **VPNs (Virtual Private Networks)** are used.

This section covers:

1. Site-to-Site VPNs using IPsec
2. Remote-Access VPNs using TLS

Site-to-Site VPN (IPsec)

A site-to-site VPN is a VPN between two devices and is used to connect two sites securely over the Internet.

Basic Concept



Office A and Office B are connected to the Internet.

A site-to-site VPN is established between them so devices in both offices can communicate securely.

How an IPsec Tunnel Works

A VPN tunnel is created by:

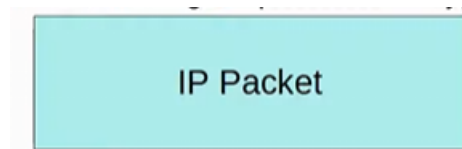
1. Encrypting the original IP packet.
2. Adding a VPN header.
3. Adding a new IP header.
4. Sending the new packet across the Internet.

When using IPsec, the original packet is encrypted before encapsulation.

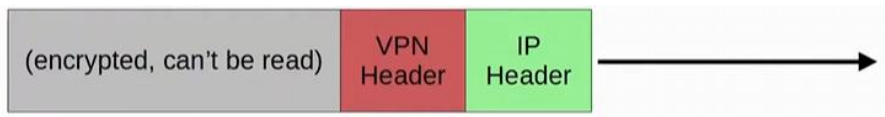
This encryption provides security.

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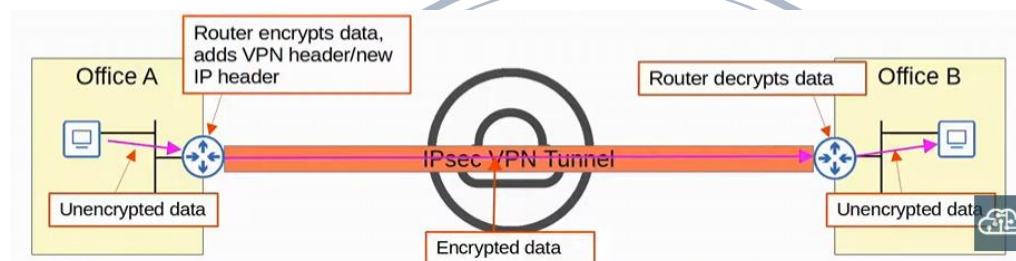
IPsec Packet Structure



After encryption and encapsulation:



IPsec Tunnel Operation



Process

1. PC in Office A sends unencrypted traffic to its default gateway.
2. Router encrypts the data.
3. Router adds:
 - IPsec VPN header
 - New IP header
4. Encrypted packet travels through the Internet.
5. Receiving router decrypts the packet.
6. Original packet is forwarded to the destination PC.

Simplified IPsec Process Summary

When a router receives traffic destined for the VPN:

1. Original packet and session key (encryption key) are combined.
2. An encryption formula is applied.
3. The encrypted packet is encapsulated with: VPN header, New IP header
4. The packet is sent through the tunnel.
5. Receiving router decrypts the packet.
6. Original packet is forwarded to its destination.

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Tunnel Endpoints

In a site-to-site VPN:

- The tunnel exists only between the two VPN endpoints.
- Typically these endpoints are routers or firewalls.

Other devices do not create VPNs themselves.

Instead:

- End devices send normal unencrypted traffic to their router.
- Router handles encryption and tunnel forwarding.

Limitations of Standard IPsec

Two limitations are highlighted.

1. No Broadcast or Multicast Support

IPsec supports: Unicast traffic

IPsec does not support:

- Broadcast traffic
- Multicast traffic

Because routing protocols such as OSPF rely on multicast traffic, they cannot operate directly across standard IPsec tunnels.

2. Full-Mesh Configuration Complexity

Large networks may require many tunnels.

Creating a full mesh manually:

- Takes time
- Requires careful planning
- Becomes labor intensive

Two solutions are introduced:

- GRE over IPsec
- DMVPN

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GRE over IPsec

GRE stands for: **Generic Routing Encapsulation**

GRE creates tunnels but does not encrypt traffic.

Therefore, GRE by itself is not secure.

However, GRE can encapsulate:

- Many Layer 3 protocols
- Broadcast traffic
- Multicast traffic

GRE provides flexibility, while IPsec provides security.

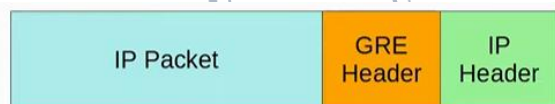
Combining them gives both advantages.

GRE Packet

Original packet:

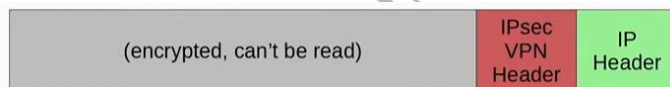


After GRE encapsulation:



GRE over IPsec Packet

The GRE packet is then encrypted and encapsulated by IPsec.



Process

1. Original packet receives:
 - GRE header
 - New IP header
2. Entire GRE packet is encrypted.
3. IPsec VPN header and another IP header are added.

Result: GRE flexibility and IPsec security

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DMVPN

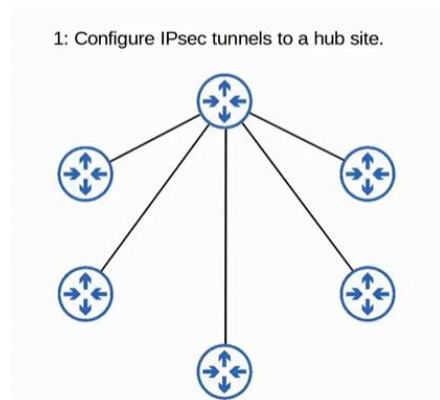
DMVPN stands for: **Dynamic Multipoint VPN**

It is a Cisco-developed solution.

DMVPN allows routers to dynamically create a full mesh of IPsec tunnels without manually configuring every tunnel.

DMVPN Step 1

Configure tunnels only to a hub router.



Spoke routers connect only to Hub

Each spoke router has:

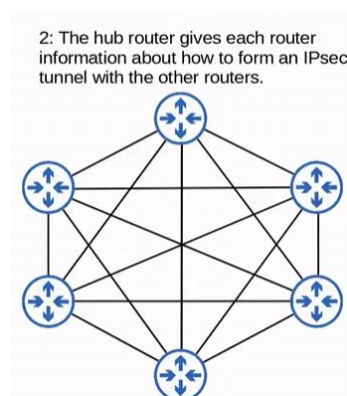
- One tunnel
- Connected to the hub

No spoke-to-spoke tunnels are manually configured.

DMVPN Step 2

The hub provides information about the other routers.

Routers then automatically create tunnels to one another.



Dynamic Full Mesh

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DMVPN Summary

DMVPN provides:

Configuration Simplicity

Hub-and-spoke style configuration.

Each router only requires one configured tunnel.

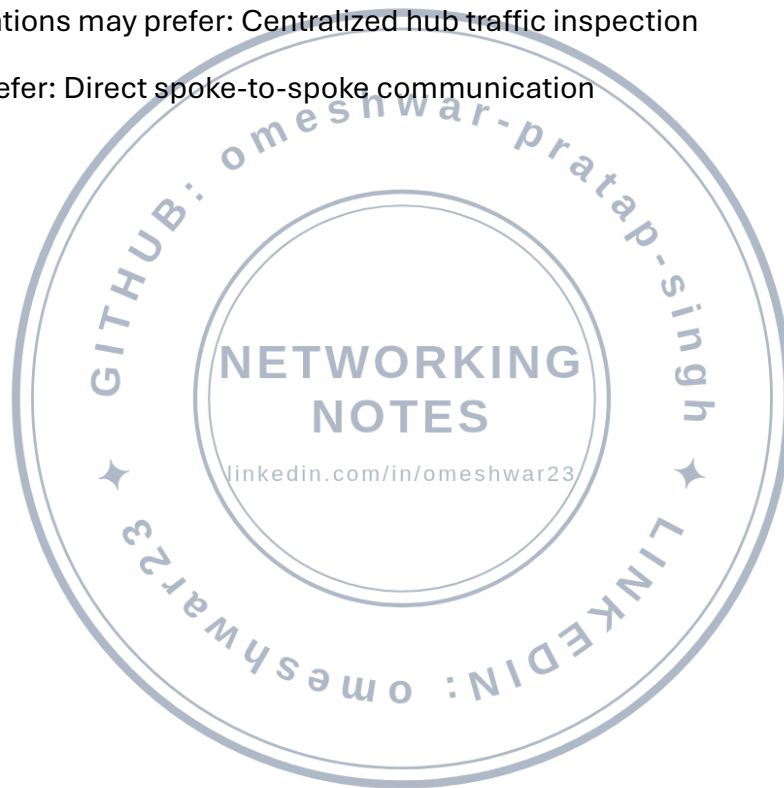
Communication Efficiency

Spoke routers communicate directly.

Traffic does not need to pass through the hub.

Some organizations may prefer: Centralized hub traffic inspection

Others may prefer: Direct spoke-to-spoke communication



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Remote-Access VPNs

Site-to-site VPNs connect sites.

Remote-access VPNs connect individual end devices.

These VPNs allow users to securely access company resources through the Internet.

TLS-Based VPNs

Remote-access VPNs typically use:

TLS (Transport Layer Security)

Site-to-site VPNs typically use:

IPsec

TLS is also the security technology used by:

HTTPS (HTTP Secure)

TLS Background

TLS was previously known as:

SSL (Secure Sockets Layer)

SSL was developed by Netscape.

When standardized by the IETF, it was renamed TLS.

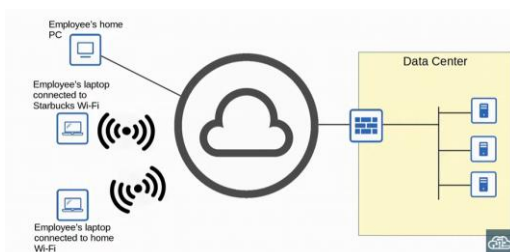
VPN Client Software

Remote-access VPNs use VPN client software.

Example: Cisco AnyConnect

VPN software is installed on end devices such as company laptops.

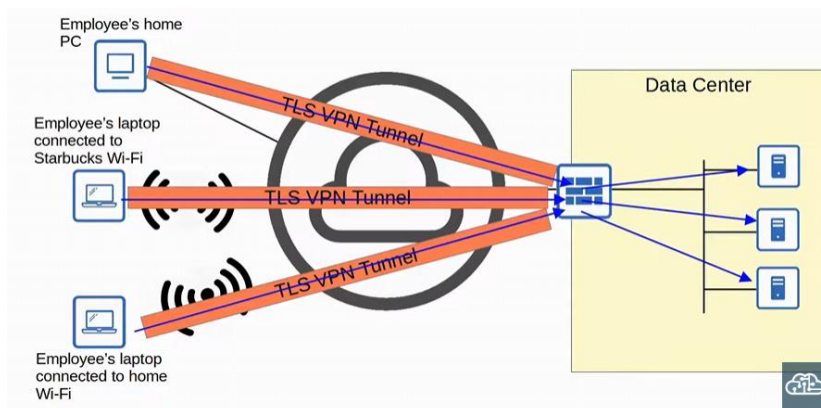
Remote Access Scenario



Users want to securely access company resources located in the data center.

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TLS VPN Tunnels: Each device forms its own TLS VPN tunnel.



The devices securely communicate with internal company servers through their VPN tunnels.

Like IPsec, TLS encrypts traffic and adds additional headers.

Site-to-Site VPN vs Remote-Access VPN

Site-to-Site VPN

Typically uses IPsec

Connects two sites

Tunnel exists between routers/firewalls

One tunnel protects many hosts

Usually permanent connectivity between sites

Used for secure site-to-site communication over the Internet

Remote-Access VPN

Typically uses TLS

Connects one device to a site

Tunnel exists between client device and company gateway

One tunnel serves one device

Usually on-demand connectivity

Used for secure access to company resources from external networks